

README.md

Differential Gene Expression Analysis Pipeline

Author: Puja Sanjay Thorat

Language: R (Version 4.x)

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Script Type: Full bioinformatics analysis workflow

Purpose of This Script

This R script performs a complete analysis of microarray-based gene expression data. It includes:

1. Loading raw gene expression, annotation, and sample metadata files
2. Cleaning and renaming sample column identifiers
3. Merging gene expression data with gene annotation
4. Computing tumor vs. normal average expression
5. Calculating fold changes and $\log_2$ fold changes
6. Identifying significantly dysregulated genes ($|\log_2FC| > 5$)
7. Annotating genes by chromosome and expression direction
8. Exploratory Data Analysis (EDA) summaries
9. Multiple visualizations:
 - DEG histograms
 - Tumor vs. Normal chromosome distribution
 - DEG percentage bar chart
 - High-resolution heatmap

- Clustered heatmap (clustermap)

This script is intended for users performing differential expression analysis and generating publication-ready gene expression visualizations.

Required Input Files

The following files must be present in the working directory:

1. Gene_Expression_Data.xlsx

- Raw expression matrix
- First column = Probe IDs
- Remaining columns = Samples (GSM format)

2. Gene_Information.csv

- Gene annotation file
- Must contain:
 - Probe_ID
 - Symbol
 - Chromosome

3. Sample_Information.tsv

- Metadata for each sample
- Must include:
 - patient (e.g., "patient: 01")
 - group (tumor/normal)

4. R_adv_assignment.R

Required R Packages

Make sure the following packages are installed:

```
install.packages(c("readxl"))
```

Base R functions (e.g., `heatmap`, `hclust`, `dist`) are also used.

Instructions: How to Run the Script

Follow these steps to run the script on **any machine**:

1. Install R (Required Version: 4.x or higher)

Download from: <https://cran.r-project.org/>

2. Place the following files in the same directory:

Gene_Expression_Data.xlsx

Gene_Information.csv

Sample_Information.tsv

R adv assignment.R

3. Open R or RStudio

4. Set the working directory:

```
setwd("/path/to/your/folder")
```

5. Run the script:

```
source("R adv assignment.R")
```

6. Output files and results will be generated in the same directory.

Output Files Produced

The script generates multiple output datasets and figures:

Data Tables

Output File	Description
<code>fold_change_d ata</code>	Raw fold-change and log ₂ FC table
<code>fc_annotated</code>	Annotated DEGs
<code>extreme_genes</code>	List of extreme fold-change genes

Plots

File	Description
<code>heatmap_highres .png</code>	High-resolution heatmap (white → yellow → orange → red)
<i>Histogram outputs</i>	DEG distribution by chromosome (Tumor, Normal, and combined)
<i>Barplots</i>	DEG percentages up/down in tumor

Output -

Part A — Data Loading

- All datasets loaded successfully: `gene_expression`, `gene_info`, `sample_info`.

Part B — Sample Renaming

- GSM IDs cleaned and renamed to `tumor_XX` / `normal_XX`.

Part C — Merging

- Expression data merged with gene annotations.
- Tumor-only and Normal-only matrices created.

Part D — Average Expression

- Mean expression calculated for each gene in Tumor and Normal.

Part E — Fold Change

- Fold change and log₂FC computed.
- Inf/NaN values handled.

Part F — Strong DEGs ($|\log_2\text{FC}| > 5$)

- Filtered list of highly deregulated genes.
- Annotation added; missing symbols flagged.

Part G — Direction of Expression

- Each gene labeled as Higher in Tumor or Normal.
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PART 2 — EDA

Part 2A

- Summaries of Tumor/Normal averages and log₂FC.
- DEG counts and extreme DEG listings.

Part 2B

- Histogram of DEGs per chromosome.

Part 2C

- Barplot: Tumor vs Normal DEG counts by chromosome.

Part 2D

- Barplot: % DEGs upregulated vs downregulated in Tumor.

Part 2E

- Heatmap of raw gene expression (scaled).

Part 2F

- Clustermap showing grouping of Tumor vs Normal samples.

Overall Conclusion

- Tumor and Normal samples separate clearly.
- Most DEGs are **upregulated in Tumor**.
- Strong fold changes and chromosome-specific DEG patterns.
- Heatmaps confirm strong biological clustering.

Notes & Assumptions

- Probe IDs must match between *gene_expression* and *gene_info*.
- Script performs extensive QC checks at every step.
- Chromosome extraction assumes format like "3", "12q", "17p13".
- Normalization is *not* performed beyond simple row scaling for visualization.